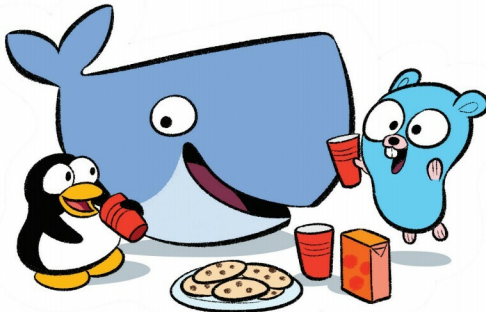


Packaging Microservices in Docker Containers

Microservices is an architectural style that structures an application as a collection of loosely coupled services. This enables the continuous delivery/deployment of large, complex applications. It also enables scaling. This tutorial explains how to deploy microservices in a Docker instance.



With rapidly changing technology, the approach towards building Web applications is changing. New styles of development are emerging and old approaches have been adopted in new ways. One such approach that has been rapidly evolving and been widely adopted is the microservices architecture. Another technology in discussion is container technology. Since the release of Docker in 2013, containers have also been adopted quickly.

What are microservices?

Unlike a giant monolithic application trying to provide all the functionality, microservices are small services running as separate processes, where each service is lined up with a specific business capability and only defines the operations necessary to that function. One big single application is broken down into a network of processes. All these services communicate with each other using a well-defined interface such as REST or message queues. Now that we know a little bit about microservices, the next question is, what do they provide in return?

Why use microservices?

Microservices make an application loosely coupled. Therefore, when in need of an upgrade/repair to a microservice, instead of building the entire application, we

can just focus on the single service that needs to be upgraded without harming any other services.

When microservices are connected through a well-defined interface, we can use different tools and languages that are best suited for the job. Scaling becomes very easy by adding copies of the microservice that is under load. Since all the microservices are independent of each other, services can continue to work in case one service fails. With microservices, we can have a small team of developers responsible for a specific task.

Why use Docker for microservices?

Docker is a container platform and is designed to simplify the task of creating and running applications by using containers. Everything required to make a piece of software run is packaged into containers. Each microservice can be packaged as a Docker container and each service instance can be deployed as a container. This containerisation of single services makes it very simple to manage and update these services. A few benefits of this approach include:

- Scaling up and down a service can simply be done by adding and removing the number of container instances.
- Containers are extremely fast to build and start. A Docker container starts in a matter of seconds.
- A container imposes limits on the CPU and memory